

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Fundamentals of Big Data (Lab)	Course Code:	
Program:	BS (Data Science)	Semester:	Spring 2023
Duration:	100 Minutes	Total Marks:	30
Paper Date:	28-March-2023	Weightage:	25
Section:	BDS-4A	Page(s):	2
Exam:	Midterm	Section:	_____
		Roll No:	_____

- Instruction/Notes:**
- You can use either jupyter notebook via anaconda prompt or Google Collab to solve the exam.
 - Properly comment your code.
 - You **CANNOT** use any built in function **NOT** taught in class.
 - If there is any ambiguity, take reasonable assumption. Questions during exam are not allowed.
 - All other rules pertaining to examinations as per NUCES policy apply.
 - Make a separate notebook to answer your questions and use the following syntax to name that notebook (21L-XXXX.ipynb)

Question 1 [Marks = 10]: Basic Python (Strings, Arrays, Loops)

You are given a text file; your task is to compute the probabilities of each word from the corpus/data. The "data.txt" can have any type of text in it, we only want to compute the probability of the words that have English alphabets only.

Probability can be computed as: $\text{Probability (words)} = \frac{\text{count (words)}}{\text{Total words}}$

Sample "data.txt": Hello to 3v3ry Body. Yours's sincerely. hello I am "Ali". I AM no body.

Total words: 13

- Implement a function *getWordsList* that receives text file name and returns a list of words after preprocessing the text file. Preprocessing means to filter the data according to the abovementioned criteria. This function opens the file, reads the data, apply filters on data and store in a list.
- Implement a function *printWord* which displays the Word, count and its probability as shown in above table (top 100 words, having higher probabilities).

To keep record of word and its count you are required to use an appropriate data structure that you have previously studied.

Workflow:

- Read the file, *getWordsList(filename)*
Apply 4 conditions
 - Convert text to lowercase (use *text.lowercase()* function)
 - Split the text into words (use *text.split(separator)* function)
 - Remove words having numeric character/s
- Get a list of words from *getWordsList* function, store the words and their counts in the appropriate data structure.
- Call *printWords* function which displays the Word, count and its probability as above table

Name: _____ Roll Number: _____ Section: _____

Word	Count	Probability	Word	Count	Probability
hello	2	2/13	i	2	2/13
to	1	1/13	am	2	2/13
body	2	2/13	ali	1	1/13
yours	1	1/13	no	1	1/13

Question - 02 [Marks = 10]: Map-Reduce

We have received the huge user comment file and we wish to perform some basic statistics on it. Write a map and reduce code to perform the following tasks **(ANY TWO)**

- Count the number of UserNames starting with each English alphabet.
- Sort the data on the basis of the length of the comment given by each user.
- Find the user who have given maximum number of comments

Sample Input File.txt:

UserName, Comment

Aliya153, Your website is superb

Sara2, You need to work on your website design

Ali45, Good !!!

Ali45, I will definitely visit again

Question - 03 [Marks=5]: K-Means Clustering

Online Retail Dataset is attached in the file.

- You need to load the dataset and preprocess it for missing values and outliers..
- You need to scale the data for better clusters.
- Split the dataset into 70:30 ratio and perform k-folds (k=5) validation.
- You are then required to perform Kmeans clustering on this dataset.
- Try with 3 and 5 numbers of clusters and visualize them using a scatter plot.
- BONUS[1 abs]:** Evaluate the model using classification matrix/ accuracy score.

Note: To score full marks, properly execute your files and display the outputs for each cell separately. Moreover, submit .ipynb file.

Good Luck!